

OJAS GRAMOPADHYE

og2186@nyu.edu +1(929)753-6104 linkedin.com/in/ojas-gramopadhye

EDUCATION

New York University, New York Aug 2024 - May 2026
Master of Science, Computer Science GPA : 3.89/4
Courses : Artificial Intelligence, Deep Learning, Robot Perception, Big Data, Algorithms, ML Sys-Ops (Spring 2026)

Indian Institute of Technology Bombay, India May 2019 - Jul 2023
Bachelor of Technology, Computer Science and Engineering GPA : 7.74/10
Courses : AI & ML, Speech Recognition, Image Processing, Database Systems, Data Structures, Software Systems

KEY PROJECTS

Flowsight : NYC Traffic Collision Analysis and Prediction Jan 2025 - May 2025
Course Project | Big Data **New York University**

- Processed millions of NYC traffic and collision records using a Kafka - MongoDB - Spark pipeline, enriching speed data with crash context through BallTree-based haversine nearest-neighbor joins
- Identified high-risk collision hotspots by training a Spark ML model, thresholding outputs into custom risk function based on incident impact, monitoring with a React dashboard and d3.js visualizations

Erica : AI Tutor agent with GraphRAG Sept 2025 - Dec 2025
Course Project | Artificial Intelligence **New York University**

- Developed a Dockerized LLM tutor agent that ingests multi-format course content, enriches it with LLM metadata, stores corpus artifacts in MongoDB, and pipelines them into GraphRAG for grounded, concept-aware retrieval
- Delivered structured, prerequisite-aware explanations by integrating semantic search, a typed knowledge graph, and Streamlit UI with subgraph visualizations, enabling the agent to answer queries with references

Multi-subject alignment : Neural Speech Decoding Sept 2025 - Ongoing
Graduate Research | NYU Video Lab **New York University**

- Improved cross-subject neural speech decoding by learning subject-invariant embeddings through contrastive alignment and adapter-based modeling across multiple ECoG participants
- Experimenting with representation learning by implementing a wav2vec2-based pretraining pipeline : masked encoder modeling, Gumbel quantization, negative sampling, and a Transformer context network for phoneme classification

WORK EXPERIENCE

Detection and Mitigation of Hallucination in Generative Models May 2023 - May 2024
Research Assistant **IBM Research | IIT Bombay**

- Improved diagnostic accuracy of LLMs to 90% by designing a Forward-Backward Chain-of-Thought evaluation pipeline and fine-tuning LLaMA-2 (7B, 70B) models for hallucination mitigation in medical QA
- Built a Subjective Medical QA benchmark and a reward-based verification pipeline that generates mixed-step reasoning signals and enables scalable, reproducible experimentation - EMNLP Findings 2024 (*arXiv: arxiv.org/abs/2403.04890*)

Japanese OCR Pipeline May 2022 - Jul 2022
Data Science Intern **Daikin**

- Delivered a full OCR pipeline for 3,000+ Japanese documents by building document-cleaning and image-normalization scripts, achieving 0.91 character-level accuracy with a MobileNetV3 + Differentiable Binarization detector
- Wrote preprocessing and evaluation pipeline to streamline dataset preparation and measure OCR performance on proprietary scanned documents

Locust Occurrence Modeling and Prediction May 2021 - Jul 2021
Data Science Associate **Dtime.ai**

- Supported an early-warning system for East African farmers by processing 40+ years geospatial vegetation and weather data from East African region into structured 10x10 square-km grids and handling class imbalance with SMOTE
- Improved predicted locust attack precision by 10% over baselines by performing pixel-level classification and modeling spatiotemporal patterns with a ConvLSTM U-Net model

TECHNICAL SKILLS

Machine Learning: LLMs, RAG systems, AI agents, NLP, deep learning, contrastive learning, representation learning, LoRA fine-tuning
Frameworks & Libraries: PyTorch, TensorFlow, HuggingFace, PyTorch Lightning, LangGraph, LangChain, SentenceTransformers, scikit-learn
Data & Systems: Apache Kafka, Apache Spark, Dask, MongoDB, PostgreSQL, ChromaDB, MLflow, Docker, Linux, Bash
Software Engineering: Python, C++, JavaScript, TypeScript, FastAPI, React, Streamlit, d3.js, Git, REST APIs